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Anonymization Techniques

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About me

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- Program Co-Director LLMs and AI Agents (Bootcamp)@HSLU
- Served as a Head of Data and AI at Witty Works.
- Built a core algorithm of Witty - inclusive writing assistant at Witty works (part of Hugging Face start-up accelerator, Finalist of Microsoft's Entrepreneurship for Positive Impact Cup)

Background:

PhD in Physics, University Grenoble Alpes. Research institutes in France, Sweden, Switzerland (Paul Scherrer Institute - ETH Domain)

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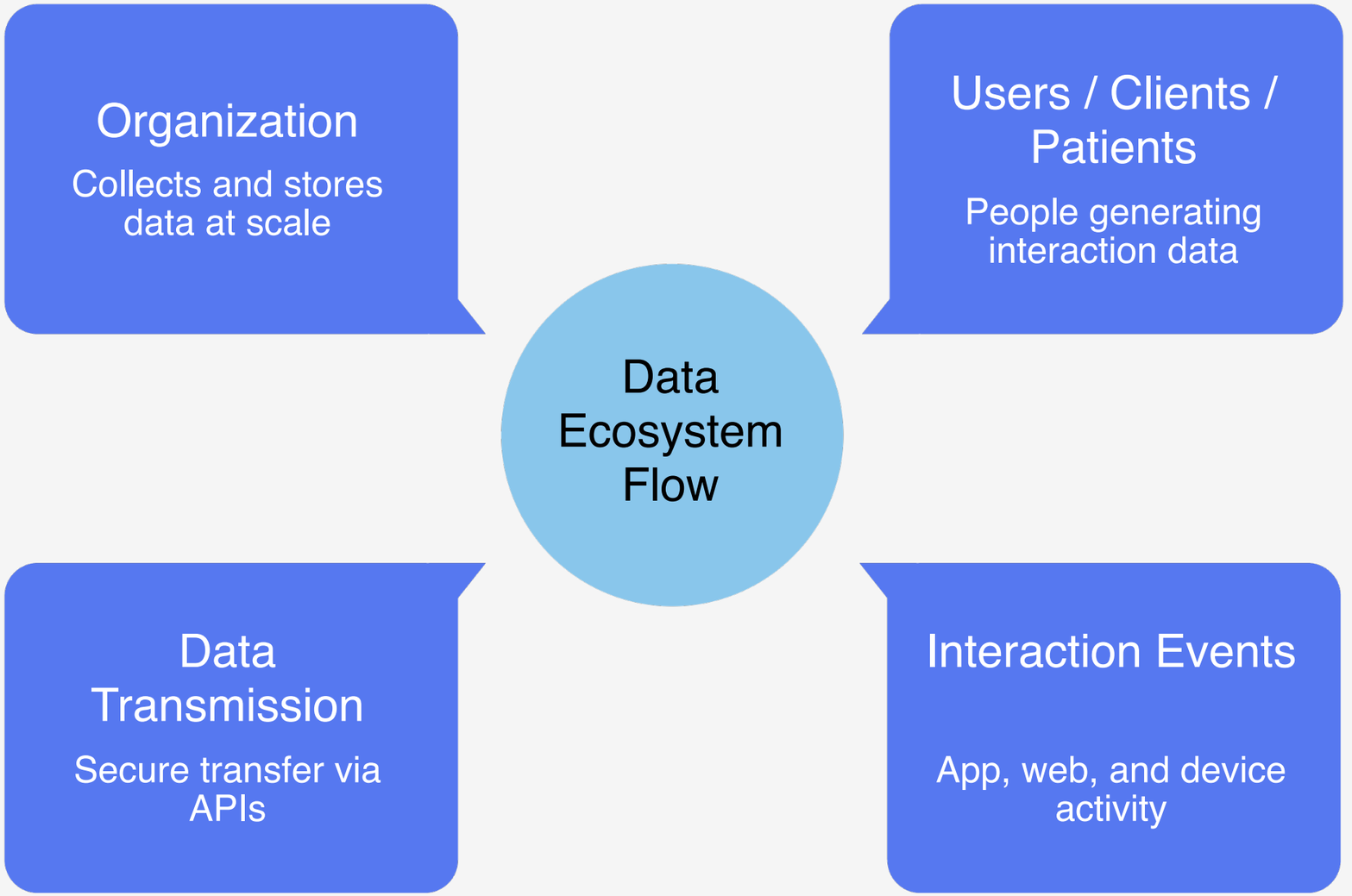
Data Anonymization: Tactics, Opportunities and Challenges

"Data that look anonymous are often not."

— Latanya Sweeney · Harvard University

The Data Ecosystem

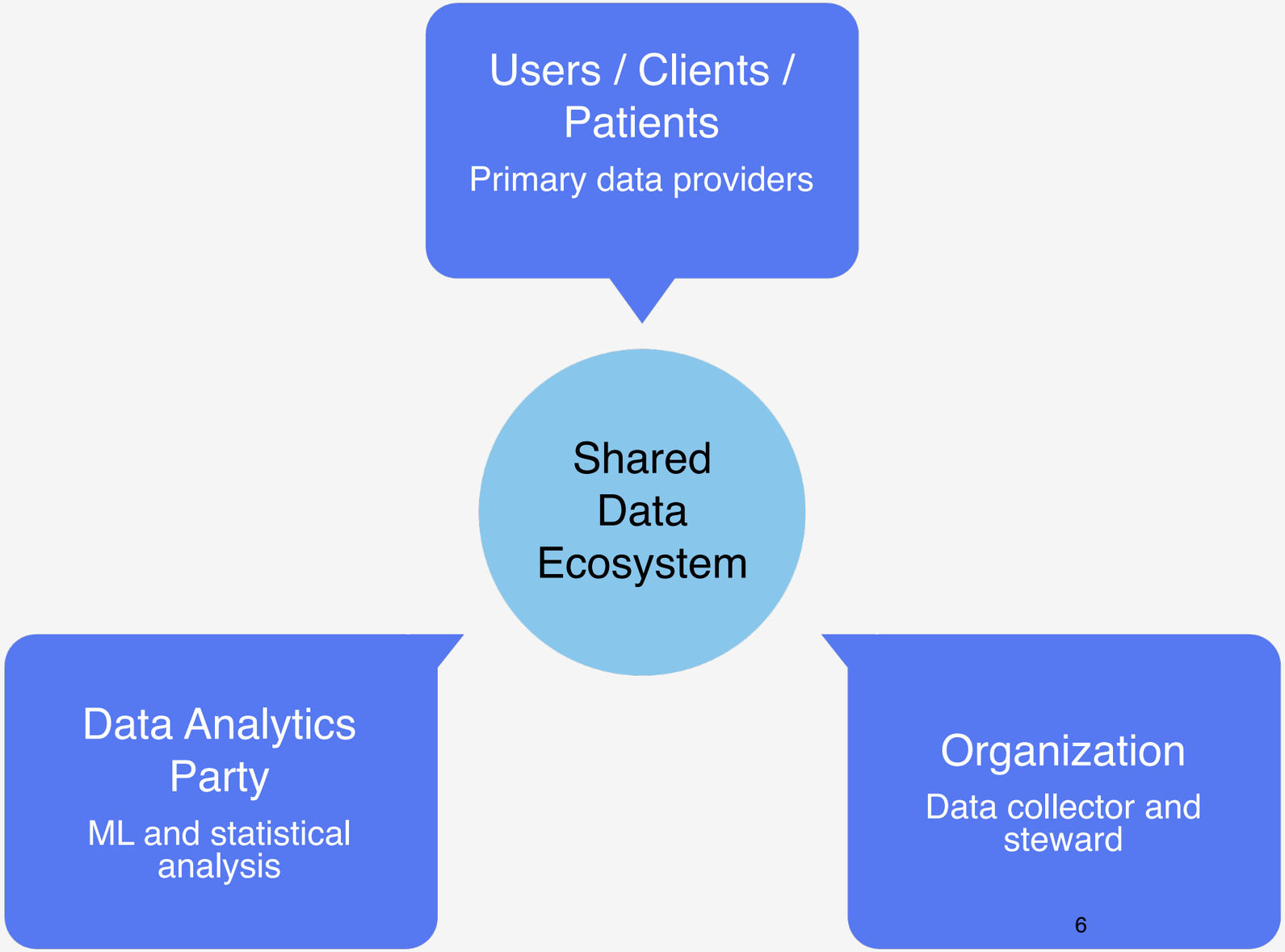
Users, clients, and patients continuously generate data — organizations collect and store it at scale.



Every interaction — from a health app to an online purchase — feeds data into organizational systems. This is the foundation of the modern data economy.

The Data Ecosystem: Adding Analytics

Organizations do not merely collect data — they share it with analytics and ML parties to extract value.



The data analytics party uses machine learning and statistical methods to derive insights from the shared dataset, creating a two-party sharing arrangement.

PART 1 — MOTIVATION

The Problems in the Data Ecosystem

The modern data ecosystem introduces four critical categories of risk that demand urgent attention.

Sharing Personal Data

Personal data is routinely shared with third parties — often beyond what users consent to or expect.

Data Retention & Secondary Use

Data collected for one purpose is retained indefinitely and repurposed for secondary uses without re-consent.

Data Breaches

Breaches expose sensitive personal data to malicious actors — with severe consequences for individuals.

Governance Questions

Who owns the data? Who is accountable? Governance frameworks struggle to keep pace with data practices.

REAL-WORLD CASE — JAN 16, 2025

Gravy Analytics Hacked, Puts Millions of Mobile Users at Risk

A major data broker breach exposed precise location histories of millions of users. Apps whose data was compromised included:

Tinder	VPN Expert
Grindr	Candy Crush
AccuWeather	Temple Run
Flightradar24	+ Thousands more

Location data linked individuals to sensitive sites — places of worship, medical clinics, political events, and private residences.

Gravy Analytics Hacked, Puts Millions of Mobile Users at Risk

Shubhangi Dua • 16 January 2025 • 2 min

351 1

A massive data location broker firm has been hacked, putting the sensitive information of millions at risk.

On January 4, 2025, Gravy Analytics, a subsidiary of Unacast, discovered that someone had unauthorized access to its AWS cloud storage area according to Unacast in an official non-compliance report to the Norwegian Data Protection Authority.

MARKET TRENDS

The Growing Demand for Data Anonymization

The primary reason for the widespread adoption of data anonymization is the expanding amount of data being gathered and stored by organizations, along with the rising demand to safeguard the privacy of individuals associated with that data.

1

Expanding Data Collection

Companies gather personal information from e-commerce platforms, governmental bodies, healthcare systems, and social media channels at unprecedented scale.

2

Rising Privacy Expectations

Increasing public concern over privacy and greater pressure to implement stricter safeguards have made data masking widely accepted across industries.

3

Global Regulatory Pressure

The expansion and tightening of global data protection regulations compel companies to implement systems that secure customers' private and sensitive information.

PART 2

A note on regulation...

Understanding the legal landscape governing data privacy and anonymization

REGULATION

GDPR — Key Provisions

Article 5(1)(e) — Storage Limitation

Personal data must be kept only for as long as necessary for the stated purpose. Retention beyond necessity is a violation.

Article 6 — Legal Basis for Data Sharing

Processing requires a legal basis: **consent**, contractual necessity, legal obligation, vital interests, public task, or legitimate interests.

Anonymization & GDPR Scope

If data is **irreversibly anonymized**, it falls **outside the scope of GDPR** entirely — a powerful incentive for proper anonymization.

Important Distinction

Pseudonymisation is NOT the same as anonymisation. Pseudonymised data is still personal data under GDPR — the original identity can potentially be restored.

REGULATION

Definition of personal data

According to the GDPR: **personal data are “any information relating to an identified or identifiable natural person”**

- **Natural person:** Data by themselves (numbers, text, pictures, audio, etc.) are not inherently personal. They only become personal when they refer to or relate to a living individual. When data refer to an organisation, deceased person, or group of individuals, they are not considered personal data under the GDPR.
- **Data are personal if they relate to an individual.** This means practically anything that someone is, has said or done, owns, may think, etc.
- **The person should be identified or identifiable.** This is the case not only through directly identifying information, such as names and contact information, but also through indirectly identifying information, for example if you can single someone out or identify them by combining datasets

REGULATION

How to assess whether data contain personal data?

- ▶ They contain directly identifying information.
- ▶ It is possible to single out an individual
- ▶ It is possible to infer information about an individual based on information in your dataset
- ▶ It is possible to link records relating to an individual.
- ▶ De-identification is still reversible.

You can assume that you are processing personal data when you collect data **directly** from people, even if the results of that collection are anonymous. But also when you use data that are **observed or derived from people**, even if those data were previously collected, made public or used for non-research purposes.

(BUT...)

REGULATION

The EU AI Act — Data Governance

The AI Act introduces new obligations around data quality and traceability — with direct implications for anonymization practices.

Article 10 — Data Quality

Training, validation, and testing datasets must be **relevant**, **representative**, **free of errors**, and **complete**.

Article 11 — Data Lineage

Organizations must trace the full data journey: from **source** through all **processing steps** to the final **output**.

Provocative Question

What is the impact of anonymization on data governance...?

Anonymization modifies data — does this break lineage? Does it affect representativeness? These tensions must be explicitly addressed.

PART 3

What are pseudonymisation and anonymisation?

Pseudonymisation

Pseudonymisation under GDPR

- Definition (Art. 4(5), Rec. 28): a safeguard that reduces linkability between data and data subjects
- Data is de-identified so it cannot lead to identification without additional information
- In theory: remove the additional info → data becomes anonymised

Common interpretation

- Replace direct identifiers (e.g., names) with pseudonyms
- Store the mapping in a separate key file, kept apart from the research data

Pseudonymous data are still **personal data** and thus subject to the GDPR. This is because the de-identification is *reversible*: identifying data subjects is still possible, just more difficult. This means that in order to use pseudonymous data, you still need to comply to all the rules in the GDPR.

PSEUDONYMIZATION

Re-identification Risks in Pseudonymized Data

1

Isolating Records

Despite pseudonymization, it may still be possible to isolate specific records as individuals remain linked to unique attributes introduced by the pseudonymization process.

2

Establishing Links

The connection between records may remain straightforward when the same pseudonymized attribute is used for a single individual, or through other shared attributes across datasets.

3

Making Inferences

Inference risks arise when an attacker can deduce the real identity of an individual from the dataset or from other databases that share the same pseudonymized attribute.

Anonymisation

Anonymisation under GDPR

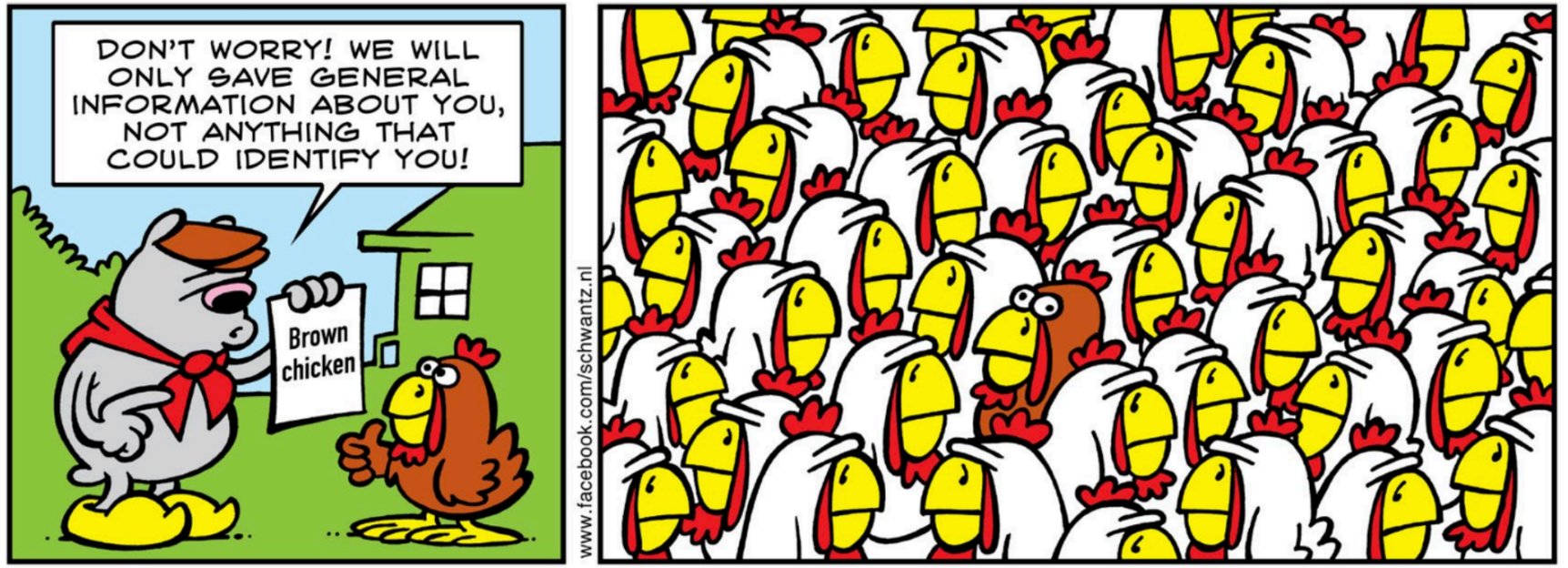
- Definition (Rec. 26): data are rendered anonymous such that the data subject is no longer identifiable
- Identification must be impossible — directly or indirectly, and by anyone (including you, the data controller)
- Once anonymised → data are no longer personal data → outside the scope of the GDPR

Important caveat ⚠️

- Everything done before anonymisation is still personal data processing
- The anonymisation process itself is also subject to GDPR
- → You still need a lawful basis, documentation, and safeguards up to the point of anonymisation

Anonymisation is very difficult to accomplish in practice!
why.

When are data anonymous?



When are data anonymous?

Basically, your data are not anonymous (personal) when they comply with any of the characteristics of personal data:

- There is directly identifiable information (e.g., name, email address, social security number, etc.).
- Data subjects can be singled out (i.e., you can tell one data subject from another within a known group of data subjects).
- It is possible to identify data subjects by linking records (“mosaic effect”), either within your own database or when using other data sources.
- It is possible to identify a data subject by inferring information about them (e.g., infer a disease by the variable “medication”), either within your own database or when using other data sources.
- It is possible to reverse the de-identification.

CORE CONCEPT

How Anonymization Works: Masking Personal Data

The process of anonymizing data generally involves methods such as masking, which hides personal markers like addresses, phone numbers, passport numbers, and Social Security IDs.

Replace Identifying Elements

Personal markers such as addresses, phone numbers, passport numbers, and Social Security IDs are hidden or replaced through encryption or random alterations.

Maintain Dataset Structure

Anonymization preserves the overall structure of the dataset, ensuring it remains usable for analysis, research, and development without exposing individual identities.

Re-identification Risk Remains

Although anonymizing data lowers the likelihood of identifying individuals, it does not guarantee full anonymity. Re-identification risks remain, particularly if anonymized data is compared with publicly available sources.

Thorough Risk Evaluation Required

Data teams must thoroughly evaluate the risks and limitations of their anonymization approaches when handling sensitive or private information.

THE BASICS

Dataset Anonymization — Intro

Consider a beer-rating dataset. The obvious first step is removing names — but is that enough?

Name	Gender	Age	Beer Rating (0–10)
Tom	M	32	8
Thomas	M	27	6
Michiel	M	24	9
Ilse	F	31	7
Ruben	M	22	10
Jonas	M	28	5

Numbers in this slide are fictional. The • Name column will be removed — but Gender + Age together may still identify individuals.

THE BASICS

Attribute Classification

Not all attributes carry the same privacy risk. We classify them into three types to guide anonymization strategy.

● Identifying Attributes

Directly identify an individual.
Example: Name, Social Security Number, Passport Number

→ **Must be removed or replaced**

● Quasi-Identifying Attributes

Can identify when combined with other data.
Example: Gender, Age, ZIP Code, Nationality

→ **Must be generalized or suppressed**

● Sensitive Attributes

The data we actually want to protect.
Example: Disease, Beer Rating, Salary, Criminal Record

→ **Must not be linkable to an individual**

⚠ Simply removing identifying attributes is **NOT sufficient** for anonymization. Quasi-identifiers must also be processed.

Why Removing Identifiers Is Not Enough — Netflix Prize (2006)

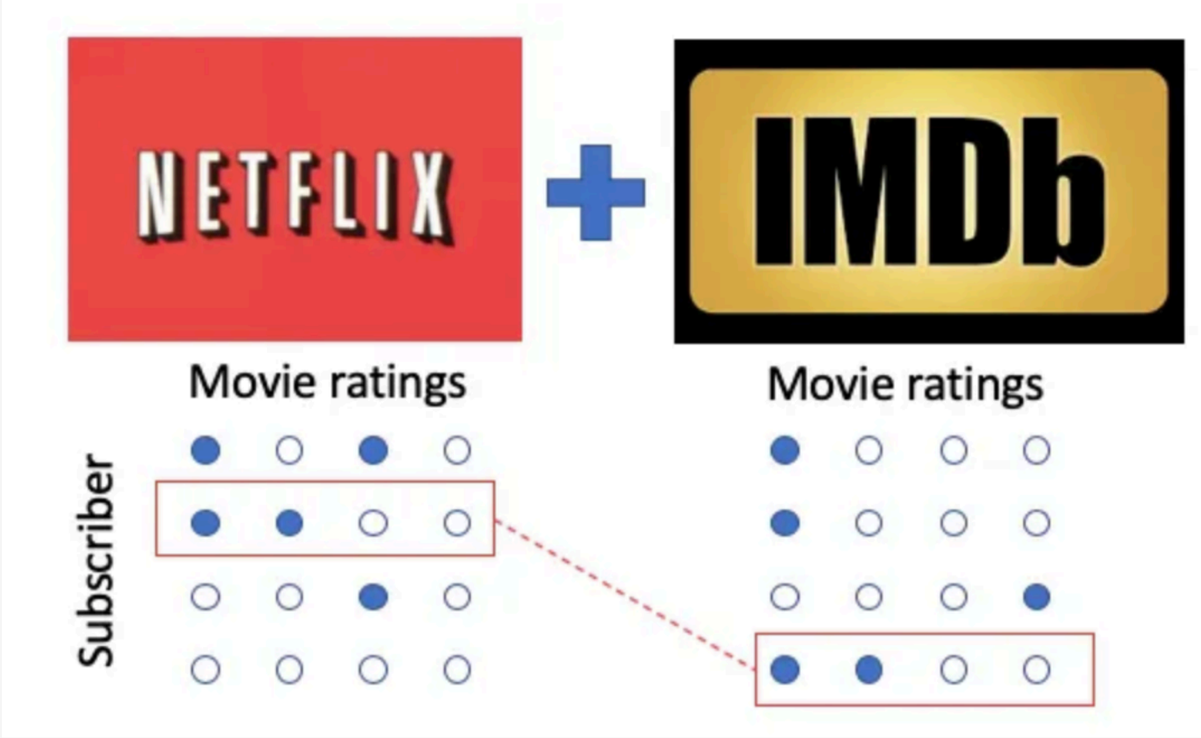
Narayanan, A. & Shmatikov, V. (2008). *"Robust de-anonymization of large sparse datasets."* IEEE Symposium on Security and Privacy.

- Oct 2, 2006: Netflix launches $1M Prize to improve its recommendation algorithm
- Released "anonymized" dataset: movie ratings from 500,000 subscribers
- Netflix's claim: all personally identifiable information (PII) removed → privacy protected ✓

- **Key question: Is removing PII enough to guarantee privacy?**

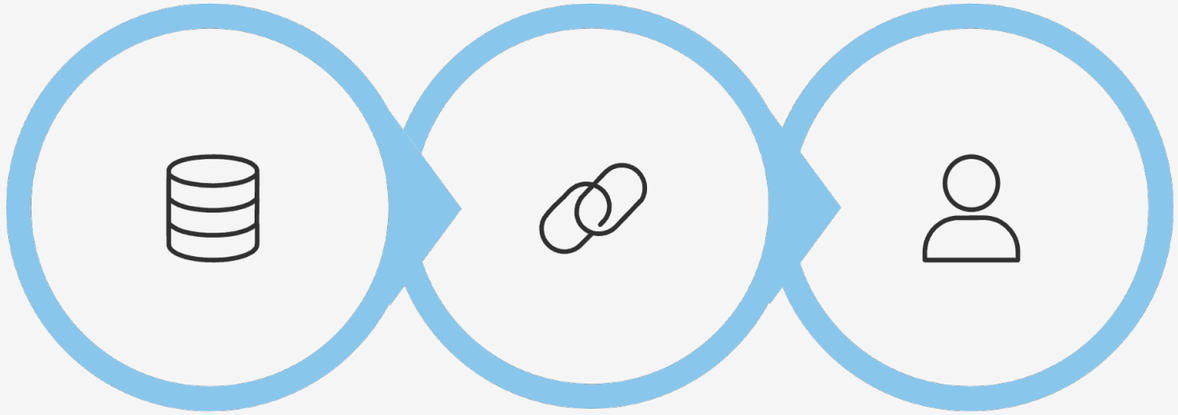
- 2007 – Narayanan & Shmatikov (UT Austin) proved the answer is no

- Takeaway: PII removal alone ≠ anonymization — auxiliary data enables re-identification



THE BASICS — CASE STUDY

Netflix De-Anonymization — How It Worked



By matching as few as 8 movie ratings with dates from the public IMDb database, Narayanan & Shmatikov could uniquely identify a Netflix subscriber with 99% confidence — even when noise had been added to the data.

This demonstrates that **quasi-identifiers in combination** — even seemingly innocuous ones like movie preferences and dates — can uniquely fingerprint individuals at scale.

Key Insight

Processing the quasi-identifiers is required for anonymous data sharing!

THE BASICS

Pseudonymization vs. Anonymization

Pseudonymization

Replace identifying fields with **pseudonyms** (artificial identifiers).

Techniques: Hashing, encryption, tokenization

Key property: **REVERSIBLE** — the original identity can be restored with the right key

Still considered **personal data** under GDPR.

Anonymization

Process data (stripping, generalizing, suppressing elements) so it **can no longer identify a person**.

Techniques: Generalization, suppression, aggregation

Key property: **IRREVERSIBLE** — re-identification should be computationally infeasible

Falls **outside GDPR scope** when done correctly.

THE BASICS

Pseudonymization — Common Techniques

Pseudonymization can be implemented with several techniques — each with different reversibility, performance, and security properties.

Hashing

One-way function producing a fixed-size output. Irreversible in theory — but brute-force and rainbow-table attacks can recover common values. Salted hashes (per-record random salt) defeat rainbow tables.

Keyed-Hash Function (HMAC)

A hash combined with a secret key. Without the key, reversing is computationally infeasible — far stronger than plain or salted hashing when the key is well-protected.

Symmetric Encryption

Reversible transformation (e.g., AES-256). Requires secure key management — key loss = data loss; key theft = full exposure. Used when linkage or restoration is needed.

Tokenization

Sensitive values are replaced with random tokens stored in a separate secure vault. Standard in PCI/finance — tokens have no mathematical relation to the original value.

PART 4

Data Anonymization — The Basics

Attribute classification, re-identification risks, and foundational techniques

THE BASICS — STEP-BY-STEP

Anonymization Step 1 — Original Dataset

Starting point: a fully labelled dataset with all three attribute types visible.

● Name (Identifying)	● Gender (QID)	● Age (QID)	● Beer Rating (Sensitive)
Tom	M	32	8
Thomas	M	27	6
Michiel	M	24	9
Ilse	F	31	7
Ruben	M	22	10
Jonas	M	28	5
Kevin	M	33	7
Jenno	F	26	8
Dairo	M	21	9

● = Identifying | ● = Quasi-Identifying | ● = Sensitive. Numbers in this slide are fictional.

THE BASICS — STEP-BY-STEP

Anonymization Step 2 — Names Removed

The identifying attribute (Name) is removed. But the dataset is **still identifiable** — quasi-identifiers remain as-is.

● Name (Removed)	● Gender (QID)	● Age (QID)	● Beer Rating (Sensitive)
—	M	32	8
—	M	27	6
—	M	24	9
—	F	31	7
—	M	22	10
—	M	28	5
—	M	33	7
—	F	26	8
—	M	21	9

⚠ The dataset remains re-identifiable! Knowing someone is a 32-year-old male with an exact beer rating of 8 may be sufficient to identify them in context.

THE BASICS — STEP-BY-STEP

Anonymization Step 3 — Age Generalized

Exact ages are replaced with **age ranges**. Gender remains visible — still not fully anonymous.

Name	● Gender (QID)	● Age Range (Generalized)	● Beer Rating (Sensitive)
—	M	[30–35[	8
—	M	[25–30[	6
—	M	[20–25[	9
—	F	[30–35[	7
—	M	[20–25[	10
—	M	[25–30[	5
—	M	[30–35[	7
—	F	[25–30[	8
—	M	[20–25[	9

Generalization reduces precision — but combining Gender + Age Range can still uniquely identify small groups. We need one more step.

THE BASICS — STEP-BY-STEP

Anonymization Step 4 — K-Anonymity (k=3) Achieved

Gender is suppressed to *. Each age range now contains at least **3 indistinguishable records** — forming equivalence classes.

Name	Gender (Suppressed)	Age Range	Beer Rating
—	*	[30–35[	8
—	*	[30–35[	7
—	*	[30–35[	7
—	*	[25–30[	6
—	*	[25–30[	5
—	*	[25–30[	8
—	*	[20–25[	9
—	*	[20–25[	10
—	*	[20–25[	9

K-Anonymity (k=3)

- 3-anonymity achieved! Each record is indistinguishable from at least 2 others sharing the same quasi-identifiers — forming 3 equivalence classes.

THE BASICS

Data Generalization in Practice — Hierarchy Trees

Generalization hierarchies define how specific values are progressively abstracted into broader categories at each level.

1

L0 — Exact Values

Age: 21, 22, 23, 24... | Gender: Male, Female

2

L1 — Small Ranges

Age: [0–5], [5–10], [10–15]… | Gender: Male, Female

3

L2 — Wider Ranges

Age: [0–10], [10–20], [20–30]… | Gender: Male, Female

4

L3 — Very Broad

Age: [0–20], [20–40]… | Gender: *

5

L4 — Full Suppression

Age: * | Gender: *

Key Principle
Moving **up the hierarchy** increases privacy but reduces utility.
The anonymization algorithm selects the **minimum generalization level** needed to achieve the target k.

PART 5

Statistical approaches to de-identification

K-anonymity, l-diversity and their limitations

TECHNIQUES

Broadening Methods

This technique involves expanding or simplifying data attributes by modifying their scope or range (for example, substituting a specific city with a larger region or replacing an exact week with a broader month).

Data Aggregation

Combining individual data points into summary statistics, removing granular personal detail.

K-anonymity

Ensuring each record is indistinguishable from at least **k-1** other records in the dataset.

L-diversity

Extending k-anonymity by ensuring sensitive attributes have diverse values within each equivalence group.

T-closeness

Requiring that the distribution of sensitive data within any group closely mirrors its distribution in the full dataset.

DEEP DIVE

K-Anonymity — Definition

> **"An individual cannot be distinguished from at least k-1 other individuals in the dataset based on their quasi-identifying attributes."**

Mechanism

Manipulate records so that groups of at least size **k** are formed, all sharing the same quasi-identifier values → **equivalence classes**

Privacy/Utility Trade-off

K ↑ → Privacy ↑ (Utility ↓) Larger k means stronger anonymity but greater information loss through generalization.

Technique 1: Generalization

Age 28 → [20–30]
Replace exact value with a range or category

Technique 2: Suppression

Name / ZIP / Rating → *
Replace value with a wildcard to hide it entirely

DEEP DIVE — K-ANONYMITY EXAMPLE

K-Anonymity Worked Example — Original Data

Dataset: 9 records with Gender, Age, Nationality, and Disease (sensitive attribute). Generalization hierarchies are available for Age, Gender, and Nationality.

Gender	Age	Nationality	Disease
M	28	Chinese	Hepatitis
F	43	American	Flu
M	37	Japanese	Cancer
F	52	Korean	Cardiac
M	31	Chinese	Flu
F	41	Canadian	Cancer
M	26	Japanese	Hepatitis
F	49	American	Flu
M	38	Korean	Cancer

Age Hierarchy

L0: exact → L1: [25–30] → L2: [25–35]→ ... → L4: *

Gender Hierarchy

L0: M/F → L1: *

Nationality Hierarchy

L0: Country → L1: Continent → L2: *

DEEP DIVE — K-ANONYMITY EXAMPLE

K-Anonymity Worked Example — Applying Generalization

We begin generalizing quasi-identifiers. Age is moved to ranges; Nationality is moved to continents.

Gender	Age Range	Region	Disease
M	[25–30]	Asia	Hepatitis
F	[40–45]	N-America	Flu
M	[35–40]	Asia	Cancer
F	[50–55]	Asia	Cardiac
M	[30–35]	Asia	Flu
F	[40–45]	N-America	Cancer
M	[25–30]	Asia	Hepatitis
F	[45–50]	N-America	Flu
M	[35–40]	Asia	Cancer

Applying generalization hierarchies:

- Chinese/Japanese/Korean → **Asia**
- American/Canadian → **N-America**
- Age exact values → **5-year ranges**

DEEP DIVE — K-ANONYMITY EXAMPLE

K-Anonymity Worked Example — Forming Equivalence Classes

Records with the same quasi-identifier combination are grouped into **equivalence classes**. Gender is further suppressed to * to achieve k=3.

Gender	Age Range	Region	Disease
*	[25–30]	Asia	Hepatitis
*	[35–40]	Asia	Cancer
*	[35–40]	Asia	Cancer
*	[40–45]	N-America	Flu
*	[40–45]	N-America	Cancer
*	[40–45]	N-America	Flu

Records that do not fit in an equivalence class of size ≥k are **dropped (suppressed)**.

Class 1

*, [25–30], Asia

Class 2

*, [35–40], Asia

Class 3

*, [40–45], N-America

DEEP DIVE — K-ANONYMITY EXAMPLE

K-Anonymity Result — Before & After

Before (Original)

Gender	Age	Nationality	Disease
M	28	Chinese	Hepatitis
M	37	Japanese	Cancer
F	43	American	Flu
F	41	Canadian	Cancer
F	43	American	Flu

→

After (3-Anonymity)

Gender	Age	Region	Disease
*	[25–30[	Asia	Hepatitis
*	[35–40[	Asia	Cancer
*	[40–45[	N-America	Flu
*	[40–45[	N-America	Cancer
*	[40–45[	N-America	Flu

3-Anonymity Achieved

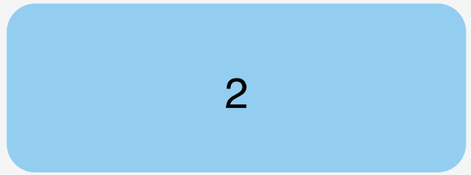
DEEP DIVE

Generalization Hierarchy — Utility Loss Visualization

As we move up the generalization hierarchy, we gain privacy but lose information quality in multiple dimensions.

L0 — Full Precision

Exact values. Maximum utility. Zero anonymization.

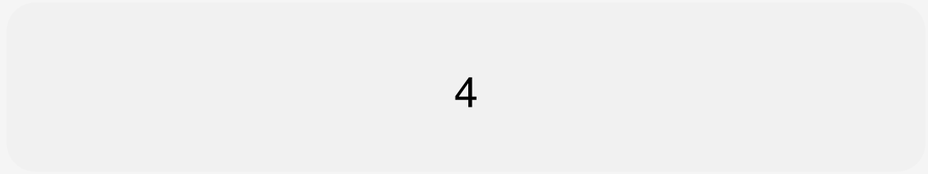


L1 — Minor Generalization

Small ranges. Slight loss of precision. Minor semantic change.

L2 — Moderate Generalization

Wider ranges. Noticeable precision loss. Meaning starts to blur.



L3 — Heavy Generalization

Very broad categories. Significant utility loss. Semantic meaning largely lost.

L4 — Full Suppression (*)

No information remains. Maximum privacy. Zero utility from this attribute.

DEEP DIVE

K-Anonymity Limitations

Even with k-anonymity achieved, two powerful attack vectors remain:

Homogeneity Attack

If all records in an equivalence class share the **same sensitive attribute value**, an attacker can infer it with certainty.

Example: All 3 records in [35–40[, Asia have Disease = **Cancer**. The attacker knows — even without identifying who is who.

Background Knowledge Attack

An attacker who knows someone is in a particular equivalence class can use background knowledge to narrow down the sensitive attribute.

Example: Knowing "Jan doesn't have heart disease" eliminates some records, revealing the probable diagnosis.

Additional Limitation K-anonymity performs poorly on **high-dimensional datasets** — with many quasi-identifiers, almost every record becomes unique, making k-anonymity impossible or extremely costly to achieve.

DEEP DIVE

L-Diversity

L-diversity was designed specifically to tackle the **homogeneity attack** that defeats k-anonymity.

Gender	Age	Region	Disease
*	[25–30]	Asia	Cardiac
*	[25–30]	Asia	Cancer
*	[25–30]	Asia	Infectious
*	[35–40]	Asia	Cardiac
*	[35–40]	Asia	Cancer
*	[35–40]	Asia	Hepatitis
*	[40–45]	N-America	Flu
*	[40–45]	N-America	Cancer
*	[40–45]	N-America	Cardiac

The Rule

Each equivalence class must contain at least **L different values** for the sensitive attribute. No single value should dominate.

Counter-Example (Fails!)

If all 3 records in a class have Disease = Cancer → l-diversity **violated**. Attacker knows the diagnosis.

DEEP DIVE

Problems with L-Diversity

L-diversity ensures value diversity, but two more subtle attacks remain possible:

Skewness Attack

H1N1 is a **rare disease**. Even if only 1 of 3 records has it, the probability for a record in that class is dramatically higher than in the general population.

L-diversity is technically met — but meaningful information is leaked.

Similarity Attack

"Dries is under 30, he has a stomach-related disease." Even with 3 diverse values (Gastritis, Gastric Ulcer, Stomach Cancer), the attacker learns the **category** of disease — which may be highly sensitive.

⚠ L-diversity treats all attribute values as equally sensitive. In reality, **semantic closeness** between values can still leak meaningful information. Enter: **T-closeness.**

DEEP DIVE

Applicability of L-Diversity

> **"Nice in theory, hard to achieve in practice."**

Harsh Generalizations Required

Meeting l-diversity often forces aggressive generalization — drastically reducing the **utility** of the anonymized dataset.

Multiple Sensitive Attributes Problem

When a dataset has **multiple sensitive attributes**, l-diversity must be satisfied for **each attribute individually**. This compounds the utility loss significantly.

DEEP DIVE

Practical vs. Theoretical Utility

Not all utility is measured the same way — the right metric depends on how the anonymized data will be used.

Practical Utility

Use-case dependent. Measures how well the anonymized data performs for its specific purpose.

Examples:

- ML model accuracy on anonymized training data
- Query answer accuracy for optimization problems
- Business decision quality based on aggregated data

Must be evaluated experimentally — no formula works universally.

Theoretical Utility

Mathematical and generic. Quantifies information loss independent of any specific use case.

Examples:

- Normalized Certainty Penalty (NCP)
- Discernibility Metric
- Average Equivalence Class Size

Enables comparison across anonymization configurations without knowing the end use.

PART 6

Data Anonymization — In Practice

Applied case study: balancing privacy, utility, and legal compliance

PART 5 — IN PRACTICE

Case Study: The Judiciary Dataset

We now apply everything we've learned to a real-world scenario involving a highly sensitive dataset.

The Data Owner

The **Judiciary** holds a dataset combining records of prosecuted criminals and law-abiding citizens. The data is highly sensitive by nature.

The Purpose

Share the data with a **data analyst** to train a machine learning model that assists police in tracking criminal behavior patterns.

The Challenge

Privacy must be absolute — **no one should be able to determine with certainty** whether a specific individual was convicted. The stakes are extremely high.

Domain Knowledge

Research indicates that owning a **luxury car** is a statistically strong indicator of criminal behavior — making it a sensitive quasi-identifier in this context.

CASE STUDY — STEP 1

Judiciary Dataset — Attribute Classification

The first step is classifying every attribute in the dataset before any anonymization begins.

Attribute	Type	Rationale
Name / ID	**Identifying**	Directly identifies the individual — must be removed
Date of Birth	**Quasi-Identifying**	Combined with other attributes, enables re-identification
ZIP Code	**Quasi-Identifying**	Geographic granularity enables linkage attacks
Luxury Car Owner	**Quasi-Identifying**	Strong predictor; may narrow down individuals significantly
Nationality	**Quasi-Identifying**	Narrows population subset
Convicted (Y/N)	**Sensitive**	The attribute we must protect — the core privacy concern

⊗ The attacker model must be documented first: Who will receive this data? What public datasets might they combine it with? What is the worst-case scenario?

CASE STUDY — STEP 2

Judiciary Dataset — Applying K-Anonymity

We apply generalization and suppression to quasi-identifiers to achieve k-anonymity. The goal: no record should stand out from at least k-1 others.

Before Anonymization

DOB	ZIP	Luxury Car	Convicted
1988-03-12	6003	Yes	Yes
1990-07-24	6004	No	No
1988-11-05	6003	Yes	Yes
1992-02-18	6005	No	No
1991-09-30	6004	Yes	No
1989-06-14	6005	No	Yes

After K-Anonymity (k=3)

DOB	ZIP	Luxury Car	Convicted
[1988–1992]	600*	Yes	Yes
[1988–1992]	600*	No	No
[1988–1992]	600*	Yes	Yes
[1988–1992]	600*	No	No
[1988–1992]	600*	Yes	No
[1988–1992]	600*	No	Yes

CASE STUDY — STEP 3

Judiciary Dataset — Adding L-Diversity (l=2)

K-anonymity is not enough. Because "Convicted" is binary, a homogeneity attack is possible. We require **l=2 diversity** within each equivalence class.

DOB	ZIP	Luxury Car	Convicted
[1988–1992]	600*	Yes	Yes
[1988–1992]	600*	Yes	No
[1988–1992]	600*	Yes	Yes
[1988–1992]	600*	No	Yes
[1988–1992]	600*	No	No
[1988–1992]	600*	No	No

L-Diversity (l=2) Met

Each equivalence class (grouped by Luxury Car = Yes / No) contains both "Convicted: Yes" and "Convicted: No" values.

No attacker can determine with certainty whether a given individual was convicted.

PART 6 — KEY TAKEAWAYS

The Anonymization Process — 4 Key Steps

A rigorous anonymization pipeline follows four sequential steps. Each step builds on the previous and directly affects the quality of the final result.

1. Attribute Categorization

Identify, quasi-identify, sensitive

2. Generalization Hierarchy Creation

Build abstraction levels per attribute

3. Privacy Model Application

Apply k-anonymity, l-diversity

4. Quality Assessment & Assurance

Validate utility and privacy guarantees

KEY TAKEAWAYS — STEP 1

Step 1: Attribute Categorization

The Three Classes

Every attribute must be labelled: **Identifying**, **Quasi-Identifying**, or **Sensitive**.

Questions to Ask

- Who are you sharing the data with?
- How sensitive is the sensitive attribute?
- What if a breach occurs? What's the damage?
- What other data is publicly available about these individuals?

Do This First!

Create an attacker model and document your assumptions before you begin anonymization.

Without a clear attacker model, you cannot determine what constitutes sufficient anonymization for your context.

KEY TAKEAWAYS — STEP 2

Step 2: Generalization Hierarchy Creation

Quality In = Quality Out

The quality of your generalization hierarchies **directly determines** the quality of your anonymized data. Poorly designed hierarchies = poor utility.

Use-Case Awareness

If the purpose of the data sharing is known, design hierarchies that **preserve the dimensions most important** for that use case.

Hierarchy Design Tips

- Give the anonymization tool enough hierarchy levels — but don't over-engineer
- ML techniques (e.g., clustering) can help discover natural groupings
- Semantic closeness matters — group by meaning, not just value ranges

Good hierarchies allow the algorithm to find the **minimal generalization** needed for the privacy target — preserving maximum utility.

KEY TAKEAWAYS — STEP 3

Step 3: Privacy Model Application

Choose Your Model

K-anonymity? L-diversity? The choice depends on the nature of the sensitive attribute and the attacker profile.

Questions to Guide the Choice

- How certain will the attacker be about the sensitive value?
- Is attribute disclosure (category-level) already problematic?
- What should absolutely never be linkable to an individual?

Practical Tip

Allow the anonymization tool to **suppress records** that cannot fit into a valid equivalence class — forced inclusion degrades privacy.

Refer Back to Your Attacker Model!

The privacy model parameters (k, l) should be calibrated directly against the attacker model you documented in Step 1.

KEY TAKEAWAYS — STEP 4

Step 4: Quality Assessment & Assurance

Theoretical Metrics

Use mathematical utility metrics (NCP, discernibility) for a generic, use-case-independent baseline assessment of information loss.

Real-Life Performance

Theoretical metrics alone are insufficient. **Experimental validation is necessary** — run the anonymized data through the actual downstream use case (e.g., train the ML model and measure accuracy).

Iterative Refinement

If the use case is known, **tweak weights during the anonymization process** to optimize for the dimensions that matter most for your application.

Quality assurance is not a one-time event. As data evolves and use cases shift, **re-assessment** of anonymization quality is essential.

SOLUTIONS

Anonymization Protects Privacy When Breaches Occur

Cyber Resilience!

Accept Breaches Will Happen

The question is not *if* a breach will occur, but *when*. Anonymization reduces the **impact** of that breach — not just the probability.

Stolen Data Cannot Harm Individuals

Properly anonymized data cannot be used to **blackmail, intimidate, or discriminate** against specific individuals — the link back to identity is broken.

Data Retains Learning Value

Even after a breach, anonymized data can **still be used for analytical and learning purposes** — the breach does not destroy its value for legitimate use.

SOLUTIONS

Anonymization Enables...

Data Collaborations

Share data with third parties — analysts, researchers, partners — without compromising individual privacy. Opens new collaborative possibilities.

Long-Term Data Retention

Anonymized data can be retained indefinitely for future learning and analysis. **GDPR storage limitations no longer apply** to truly anonymized data.

High-Quality Privacy-Friendly Analytics

Anonymized data is no longer personal data → GDPR does not apply → organizations can derive value while maintaining compliance and protecting reputation.

Reputational Protection

Avoid the reputational, financial, and legal damage of privacy incidents by ensuring sensitive data is never unnecessarily retained or shared in identifiable form.

Alternative: **Synthetic data generation** can also enable sharing — but note that privacy issues may still arise with synthetic data if not carefully validated.

CHALLENGES

Anonymization Pitfalls and Challenges

Despite its benefits, anonymization in real-world settings faces significant ongoing challenges:

Multi-Source Anonymization

Anonymizing data from **multiple sources simultaneously** is exponentially more complex. Data joined from different systems may re-enable linkage attacks even when each source is individually anonymized.

Growing Datasets Over Time

Real datasets are not static. As **new records are added** over time, previously anonymous equivalence classes may become invalid or identifiable — requiring continuous re-anonymization.

Public Data & Data Breaches

The anonymity guarantee assumes a certain attacker model. Public datasets and ongoing data breaches **continuously expand what attackers know** — eroding anonymity that was valid at the time of publication.

Traditional Anonymization vs. Synthetic Data

There is a growing debate: when is **synthetic data generation** preferable to traditional anonymization? And can a hybrid combination provide better privacy-utility trade-offs?

FUTURE DIRECTIONS

Beyond Traditional Anonymization

Emerging privacy-preserving techniques complement — and in some domains replace — generalization-based anonymization.

Homomorphic Encryption

Perform computations directly on encrypted data without decrypting. Expensive today, but maturing rapidly — relevant for cloud compute on sensitive datasets.

Federated Learning

Train models on data that stays decentralized on user devices or partner servers. Only gradients or model deltas are shared — often combined with differential privacy.

Blockchain & Decentralized Sharing

Tamper-resistant ledgers for data provenance, consent tracking, and auditability — without a single point of trust. Still experimental for production at scale.

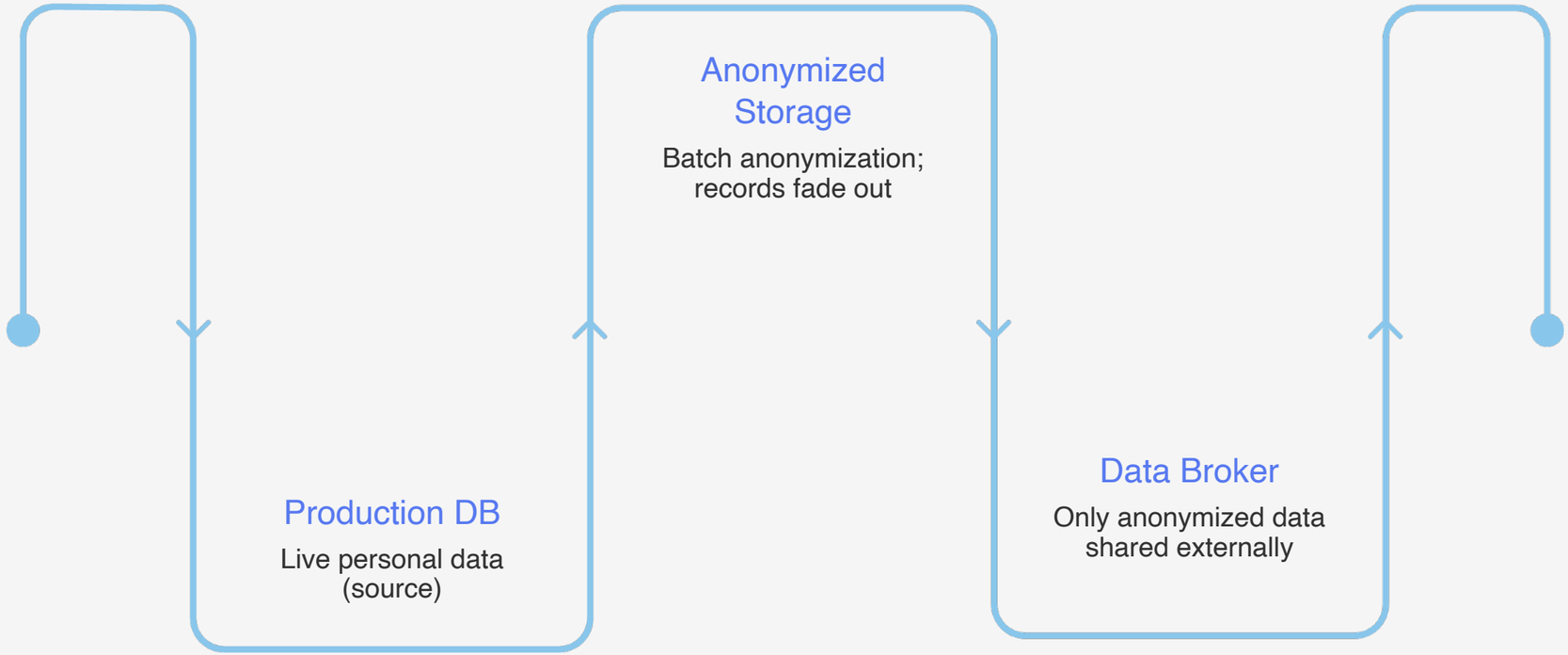
Synthetic Data & Generative Models

Model-generated records that preserve statistical properties of real data. Privacy is NOT automatic — synthetic data can leak membership or reveal outliers without careful validation.

SOLUTIONS

ML / Business Intelligence Architecture

Use anonymized data wherever possible!



Batch Anonymization

Move old records from production to long-term storage in batches. Apply anonymization at each transfer. "Fade out" data over time.

Continuous Risk Assessment

Continuously evaluate anonymization quality considering newly added data, newly available public data, and recent breaches.

SOLUTIONS

Complying with Data Governance Requirements

Anonymization must be integrated into data governance practice — not treated as an afterthought.

Data Quality Labels

Attach **meaningful utility metrics** to anonymized datasets so downstream users understand the data's quality and limitations.

Design metrics that allow quality assessment **without needing to see the original personal data** — protecting the data owner while informing the recipient.

Data Lineage

Document the complete data journey:

- Original data sources
- All processing steps applied
- Anonymization tactics used (model, parameters, suppression rate)
- Date and version of anonymization

This satisfies AI Act Article 11 requirements and enables auditability.

USE CASES

Retail, Education & Manufacturing Applications

Retail & E-commerce

Anonymize Customer Data

Purchase history, preferences, and transaction data are anonymized to generate insights into trends and behaviors without exposing sensitive information.

Regulatory Compliance

Guarantees adherence to GDPR and the California Consumer Privacy Act (CCPA), which safeguard consumer rights and personal privacy.

Education

Protect Student Information

Grades, attendance records, and personal details are anonymized to facilitate meaningful research and performance assessment while ensuring privacy.

FERPA Compliance

Aids in compliance with the Family Educational Rights and Privacy Act (FERPA), which safeguards student information.

CHALLENGES

Key Obstacles in Data Anonymization

There are several significant hurdles to effective data anonymization that organizations must navigate carefully.

Preventing Re-identification

Even with comprehensive efforts to anonymize data, the risk of re-associating data with specific individuals remains. Modern machine learning algorithms are now capable of detecting patterns within anonymized datasets, and sophisticated linkage methods allow attackers to merge various datasets more easily.

Privacy vs. Data Usability

Achieving the right balance between maintaining privacy and ensuring the usability of data is a significant challenge. Sensitive medical records typically require more rigorous anonymization compared to less sensitive demographic data.

Establishing Global Standards

As the value of data grows, the need for robust and uniform oversight of data anonymization practices has become increasingly critical. Although GDPR provides robust protection, it can make data sharing for business and research more difficult.

Leveraging AI & ML

AI/ML presents both a challenge and an opportunity. It can identify PII and generate synthetic datasets via GANs, but could also assist in de-anonymization and re-identification of individuals from previously protected datasets.

PART 7

Extras & Further Reading

Bonus research: location trace datasets and real-world privacy implications

EXTRA — BONUS RESEARCH

The Collection of Large-Scale Location Trace Datasets

Location data is among the most sensitive personal data collected at scale — revealing daily routines, relationships, health conditions, and political affiliations.

Research Paper

De Boeck K, Verdonck J, Willocx M, Naessens V.
*"Big broker is tracking you!"*ACSAC '25 — Annual Computer Security Applications Conference

Key Finding

Large-scale collection of location traces by data brokers enables inference of sensitive information about individuals — far beyond what users consent to or expect from seemingly benign apps.

Exercises